

● MEDIUM

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

03

10 questions · ~15 min

Q1

GRID-IN ADVANCED MATH

$$\frac{-54}{w} = 6$$

What is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$\frac{x + 9x - 9}{x + 9} = 7$$

What is the solution to the given equation?

- A. 7
- B. 9
- C. 16
- D. 63

The speed of sound in dry air, v , can be modeled by the formula $v = 331.3 + 0.606T$, where T is the temperature in degrees Celsius and v is measured in meters per second. Which of the following correctly expresses T in terms of v ?

A. $T = \frac{v + 0.606}{331.3}$

B. $T = \frac{v - 0.606}{331.3}$

C. $T = \frac{v + 331.3}{0.606}$

D. $T = \frac{v - 331.3}{0.606}$

An oceanographer uses the equation $s = \frac{3}{2}p$ to model the speed s , in knots, of an ocean wave, where p represents the period of the wave, in seconds. Which of the following represents the period of the wave in terms of the speed of the wave?

- A. $p = \frac{2}{3}s$
- B. $p = \frac{3}{2}s$
- C. $p = \frac{2}{3} + s$
- D. $p = \frac{3}{2} + s$

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

A. $x = \frac{b - 72}{y}$

B. $x = by - 72$

C. $x = \frac{by - 72}{y}$

D. $x = by - 72y$

$$x^2 = -841$$

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Q7

GRID-IN

ADVANCED MATH

In the xy -plane, what is the y -coordinate of the point of intersection of the graphs of $y = (x-1)^2$ and $y = 2x - 3$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN

ADVANCED MATH

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

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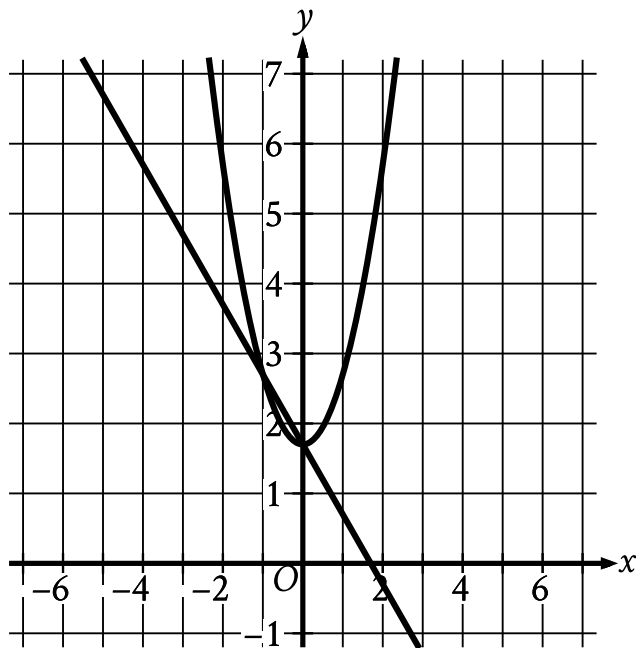
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = x^2 + 1.7$$

$$y = 1.7 - x$$

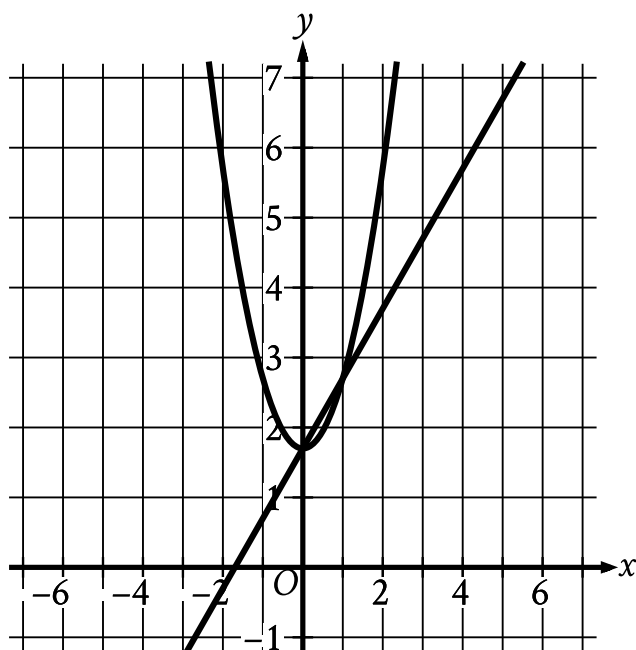
Which graph represents the given system of equations?



A.

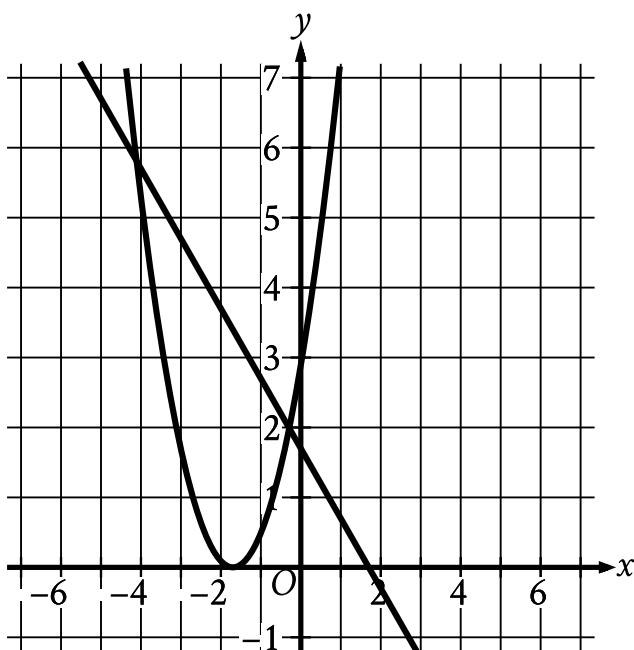
- For the line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (negative 1 comma 2.7)
 - (0 comma 1.7)
- For the parabola in the system:
 - The parabola opens upward.
 - The vertex is at point (0 comma 1.7).
 - The parabola passes through the following points:
 - (negative 1 comma 2.7)
 - (0 comma 1.7)
 - (1 comma 2.7)

B.



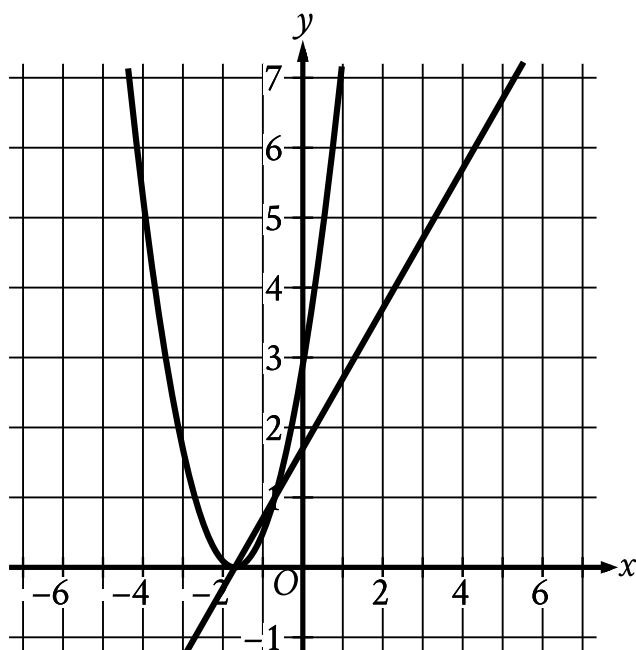
- For the line in the system:
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (0 comma 1.7)
 - (1 comma 2.7)
- For the parabola in the system:
 - The parabola opens upward.
 - The vertex is at point (0 comma 1.7).
 - The parabola passes through the following points:
 - (negative 1 comma 2.7)
 - (0 comma 1.7)
 - (1 comma 2.7)

C.



- For the line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (negative 1 comma 2.7)
 - (0 comma 1.7)
- For the parabola in the system:
 - The parabola opens upward.
 - The vertex is at point (negative 1.7 comma 0).
 - The parabola passes through the following points:
 - (negative 2.7 comma 1)
 - (negative 1.7 comma 0)
 - (negative 0.7 comma 1)

D.



- For the line in the system:
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (0 comma 1.7)
 - (1 comma 2.7)
- For the parabola in the system:
 - The parabola opens upward.
 - The vertex is at point (negative 1.7 comma 0).
 - The parabola passes through the following points:
 - (negative 2.7 comma 1)
 - (negative 1.7 comma 0)
 - (negative 0.7 comma 1)

$$x^2 - x - 1 = 0$$

What values satisfy the equation above?

A. $x = 1$ and $x = 2$

B. $x = -\frac{1}{2}$ and $x = \frac{3}{2}$

C. $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$

D. $x = \frac{-1+\sqrt{5}}{2}$ and $x = \frac{-1-\sqrt{5}}{2}$